

SQL Interview Questions with Queries

♦ Salary & Employee Queries

1. Write a query to find the second highest salary.

```
SELECT MAX(salary) AS SecondHighestSalary
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

2. Write a query to find the Nth highest salary.

```
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET N-1;
```

3. Write a query to find employees earning more than their manager.

```
SELECT e.name AS Employee, e.salary, m.name AS Manager, m.salary AS
ManagerSalary
FROM employees e
JOIN employees m ON e.manager_id = m.id
WHERE e.salary > m.salary;
```

4. Write a query to find employees with the same salary as their manager.

```
SELECT e.name AS Employee, e.salary, m.name AS Manager
FROM employees e
JOIN employees m ON e.manager_id = m.id
WHERE e.salary = m.salary;
```

5. Write a query to find employees earning above the company average salary.

```
SELECT name, salary
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees)
ORDER BY salary DESC;
```

◆ Aggregations & Grouping

6. Write a query to find duplicate records.

```
SELECT name, COUNT(*)  
FROM employees  
GROUP BY name  
HAVING COUNT(*) > 1;
```

7. Write a query to count employees per department having more than 5 employees.

```
SELECT department_id, COUNT(*) AS num_employees  
FROM employees  
GROUP BY department_id  
HAVING COUNT(*) > 5;
```

8. Write a query to perform conditional aggregation (male/female count).

```
SELECT department_id,  
COUNT(CASE WHEN gender = 'M' THEN 1 END) AS male_count,  
COUNT(CASE WHEN gender = 'F' THEN 1 END) AS female_count  
FROM employees  
GROUP BY department_id;
```

9. Write a query to count employees per job title.

```
SELECT job_title, COUNT(*) AS num_employees  
FROM employees  
GROUP BY job_title;
```

◆ Window Functions

10. Write a query to calculate running total of salaries by department.

```
SELECT name, department_id, salary,  
SUM(salary) OVER (PARTITION BY department_id ORDER BY id) AS running_total  
FROM employees;
```

11. Write a query to calculate salary difference using LAG.

```
SELECT name, salary,  
salary - LAG(salary) OVER (ORDER BY id) AS salary_diff  
FROM employees;
```

12. Write a query to rank employees by salary.

```
SELECT name, salary,  
RANK() OVER (ORDER BY salary DESC) AS salary_rank  
FROM employees;
```

13. Write a query to list top 5 highest-paid employees per department.

```
SELECT *  
FROM (  
    SELECT e.*, ROW_NUMBER() OVER (PARTITION BY department_id ORDER BY salary  
DESC) AS rn  
    FROM employees e  
) sub  
WHERE rn <= 5;
```

◆ Recursive Queries**14. Write a recursive query to find the full reporting chain.**

```
WITH RECURSIVE reporting_chain AS (  
    SELECT id, name, manager_id, 1 AS level  
    FROM employees  
    WHERE manager_id IS NULL  
    UNION ALL  
    SELECT e.id, e.name, e.manager_id, rc.level + 1  
    FROM employees e  
    JOIN reporting_chain rc ON e.manager_id = rc.id  
)  
SELECT * FROM reporting_chain ORDER BY level, id;
```

15. Write a recursive query to detect circular references in hierarchy.

```

WITH RECURSIVE mgr_path (id, manager_id, path) AS (
  SELECT id, manager_id, ARRAY[id]
  FROM employees
  WHERE manager_id IS NOT NULL
  UNION ALL
  SELECT e.id, e.manager_id, path || e.id
  FROM employees e
  JOIN mgr_path mp ON e.manager_id = mp.id
  WHERE NOT e.id = ANY(path)
)
SELECT DISTINCT id
FROM mgr_path
WHERE id = ANY(path);

```

◆ Sales & Customer Queries

16. Write a query to calculate running total of sales per customer.

```

SELECT customer_id, sale_date, amount,
SUM(amount) OVER (PARTITION BY customer_id ORDER BY sale_date) AS running_total
FROM sales;

```

17. Write a query to calculate percentage change in monthly sales.

```

SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
  SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
  AS total_sales
  FROM sales
  GROUP BY product_id, sale_month
) monthly_sales;

```

18. Write a query to find top 3 products with highest sales each month.

```
WITH monthly_product_sales AS (  
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS  
    total_sales  
    FROM sales  
    GROUP BY product_id, month  
) ,  
ranked_sales AS (  
    SELECT *, RANK() OVER (PARTITION BY month ORDER BY total_sales DESC) AS  
    sales_rank  
    FROM monthly_product_sales  
)  
SELECT product_id, month, total_sales  
FROM ranked_sales  
WHERE sales_rank <= 3  
ORDER BY month, sales_rank;
```

♦ Salary & Employee Queries

19. Write a query to find employees with salary greater than average salary in the company, ordered by salary descending.

```
SELECT name, salary  
FROM employees  
WHERE salary > (SELECT AVG(salary) FROM employees)  
ORDER BY salary DESC;
```

20. Write a query to aggregate employee names in a department as JSON array.

```
SELECT department_id, JSON_AGG(name) AS employee_names  
FROM employees  
GROUP BY department_id;
```

21. Write a query to get the first and last purchase date for each customer.

```
SELECT customer_id,  
MIN(purchase_date) AS first_purchase,  
MAX(purchase_date) AS last_purchase  
FROM sales  
GROUP BY customer_id;
```

22. Write a query to find departments with the highest average salary.

```
WITH avg_salaries AS (  
    SELECT department_id, AVG(salary) AS avg_salary  
    FROM employees  
    GROUP BY department_id  
)  
SELECT *  
FROM avg_salaries  
WHERE avg_salary = (SELECT MAX(avg_salary) FROM avg_salaries);
```

23. Write a query to find employees without a department assigned.

```
SELECT *  
FROM employees  
WHERE department_id IS NULL;
```

24. Write a query to calculate the difference in days between two dates in the same table.

```
SELECT id, DATEDIFF(day, start_date, end_date) AS days_difference  
FROM projects;
```

25. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,  
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)  
AS moving_avg_salary  
FROM employees;
```

26. Write a query to find the most recent purchase per customer using window functions.

```
SELECT *  
FROM (  
    SELECT *, ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY purchase_date  
    DESC) AS rn  
    FROM sales  
) sub  
WHERE rn = 1;
```

27. Write a query to detect hierarchical depth of each employee.

```
WITH RECURSIVE employee_depth AS (
  SELECT id, name, manager_id, 1 AS depth
  FROM employees
  WHERE manager_id IS NULL
  UNION ALL
  SELECT e.id, e.name, e.manager_id, ed.depth + 1
  FROM employees e
  JOIN employee_depth ed ON e.manager_id = ed.id
)
SELECT * FROM employee_depth;
```

28. Write a query to perform a self-join to find pairs of employees in the same department.

```
SELECT e1.name AS Employee1, e2.name AS Employee2, e1.department_id
FROM employees e1
JOIN employees e2 ON e1.department_id = e2.department_id AND e1.id < e2.id;
```

29. Write a query to pivot rows into columns (simulate fixed values).

```
SELECT department_id,
SUM(CASE WHEN job_title = 'Manager' THEN 1 ELSE 0 END) AS Managers,
SUM(CASE WHEN job_title = 'Developer' THEN 1 ELSE 0 END) AS Developers,
SUM(CASE WHEN job_title = 'Tester' THEN 1 ELSE 0 END) AS Testers
FROM employees
GROUP BY department_id;
```

30. Write a query to find customers who made purchases in every category.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM sales);
```

31. Write a query to identify employees who haven't received a raise in more than a year.

```
SELECT e.name
FROM employees e
JOIN salary_history sh ON e.id = sh.employee_id
GROUP BY e.id, e.name
HAVING MAX(sh.raise_date) < CURRENT_DATE - INTERVAL '1 year';
```


32. Write a query to rank salespeople by monthly sales, resetting rank every month.

```
SELECT salesperson_id, sale_month, total_sales,  
RANK() OVER (PARTITION BY sale_month ORDER BY total_sales DESC) AS monthly_rank  
FROM (  
    SELECT salesperson_id, DATE_TRUNC('month', sale_date) AS sale_month,  
    SUM(amount) AS total_sales  
    FROM sales  
    GROUP BY salesperson_id, sale_month  
) monthly_sales;
```

33. Write a query to calculate percentage change in sales compared to previous month.

```
SELECT product_id, sale_month, total_sales,  
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY  
sale_month)) * 100.0 /  
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS  
pct_change  
FROM (  
    SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)  
    AS total_sales  
    FROM sales  
    GROUP BY product_id, sale_month  
) monthly_sales;
```

34. Write a query to retrieve last 5 orders for each customer.

```
SELECT *  
FROM (  
    SELECT o.*, ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date  
DESC) AS rn  
    FROM orders o  
) sub  
WHERE rn <= 5;
```

35. Write a query to find employees with no salary changes in last 2 years.

```
SELECT e.*  
FROM employees e  
LEFT JOIN salary_history sh ON e.id = sh.employee_id AND sh.change_date >=  
CURRENT_DATE - INTERVAL '2 years'  
WHERE sh.employee_id IS NULL;
```

36. Write a query to find department with lowest average salary.


```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
ORDER BY avg_salary
LIMIT 1;
```

37. Write a query to list employees whose names start and end with same letter.

```
SELECT *
FROM employees
WHERE LEFT(name, 1) = RIGHT(name, 1);
```

38. Write a query to detect circular references in employee-manager hierarchy.

```
WITH RECURSIVE mgr_path (id, manager_id, path) AS (
  SELECT id, manager_id, ARRAY[id]
  FROM employees
  WHERE manager_id IS NOT NULL
  UNION ALL
  SELECT e.id, e.manager_id, path || e.id
  FROM employees e
  JOIN mgr_path mp ON e.manager_id = mp.id
  WHERE NOT e.id = ANY(path)
)
SELECT DISTINCT id
FROM mgr_path
WHERE id = ANY(path);
```

39. Write a query to get running total of sales per customer.

```
SELECT customer_id, sale_date, amount,
SUM(amount) OVER (PARTITION BY customer_id ORDER BY sale_date) AS running_total
FROM sales;
```

40. Write a query to find department-wise 90th percentile salary.

```
SELECT department_id, salary,
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY
department_id) AS pct_90_salary
FROM employees;
```

41. Write a query to find employees whose salary is a prime number.

```

WITH primes AS (
  SELECT generate_series(2, (SELECT MAX(salary) FROM employees)) AS num
  EXCEPT
  SELECT num FROM (
    SELECT num, UNNEST(ARRAY(
      SELECT generate_series(2, FLOOR(SQRT(num))) AS divisor
    )) AS divisor
    WHERE num % divisor = 0
  ) composite
)
SELECT *
FROM employees
WHERE salary IN (SELECT num FROM primes);

```

42. Write a query to find employees who worked in multiple departments.

```

SELECT employee_id
FROM employee_department_history
GROUP BY employee_id
HAVING COUNT(DISTINCT department_id) > 1;

```

43. Write a query to calculate difference between current and previous sales partitioned by product.

```

SELECT product_id, sale_date, amount,
amount - LAG(amount) OVER (PARTITION BY product_id ORDER BY sale_date) AS
sales_diff
FROM sales;

```

44. Write a query to find employees at lowest level in hierarchy (no subordinates).

```

SELECT *
FROM employees e
WHERE NOT EXISTS (SELECT 1 FROM employees sub WHERE sub.manager_id = e.id);

```

45. Write a query to find average order value per month and category.

```

SELECT DATE_TRUNC('month', order_date) AS order_month, category_id,
AVG(order_value) AS avg_order_value
FROM orders
GROUP BY order_month, category_id;

```

46. Write a query to create running count of employees joined each year.

```

SELECT join_year, COUNT(*) AS yearly_hires,
SUM(COUNT(*)) OVER (ORDER BY join_year) AS running_total_hires
FROM (
    SELECT EXTRACT(YEAR FROM hire_date) AS join_year
    FROM employees
) sub
GROUP BY join_year
ORDER BY join_year;

```

◆ Ranking & Percentiles

47. Write a query to rank employees by salary within their department and calculate percent rank.

```

SELECT name, department_id, salary,
RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS salary_rank,
PERCENT_RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS
salary_percent_rank
FROM employees;

```

48. Write a query to find products that have never been sold.

```

SELECT p.product_id, p.product_name
FROM products p
LEFT JOIN sales s ON p.product_id = s.product_id
WHERE s.sale_id IS NULL;

```

49. Write a query to find consecutive days where sales were above a threshold.

```

WITH flagged_sales AS (
    SELECT sale_date, amount,
    CASE WHEN amount > 1000 THEN 1 ELSE 0 END AS flag
    FROM sales
),
groups AS (
    SELECT sale_date, amount, flag,
    sale_date - INTERVAL '1 day' * ROW_NUMBER() OVER (ORDER BY sale_date) AS grp
    FROM flagged_sales
    WHERE flag = 1
)
SELECT MIN(sale_date) AS start_date, MAX(sale_date) AS end_date, COUNT(*) AS
consecutive_days
FROM groups
GROUP BY grp
ORDER BY consecutive_days DESC;

```

50. Write a query to concatenate employee names in each department.

```
SELECT department_id, STRING_AGG(name, ',') AS employee_names
FROM employees
GROUP BY department_id;
```

51. Write a query to find employees whose salary is above their department's average but below company-wide average.

```
SELECT *
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id =
e.department_id)
AND salary < (SELECT AVG(salary) FROM employees);
```

52. Write a query to list customers who purchased all products in a specific category.

```
SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);
```

53. Write a query to find employees with no entries in salary history.

```
SELECT e.*
FROM employees e
LEFT JOIN salary_history sh ON e.id = sh.employee_id
WHERE sh.employee_id IS NULL;
```

54. Write a query to show department with highest number of employees.

```
SELECT department_id, COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
ORDER BY employee_count DESC
LIMIT 1;
```

55. Write a recursive query to list all ancestors (managers) of a given employee.

```
WITH RECURSIVE ancestors AS (  
  SELECT id, name, manager_id  
  FROM employees  
  WHERE id = 123  
  UNION ALL  
  SELECT e.id, e.name, e.manager_id  
  FROM employees e  
  JOIN ancestors a ON e.id = a.manager_id  
)  
SELECT * FROM ancestors WHERE id != 123;
```

56. Write a query to calculate median salary by department using window functions.

```
SELECT DISTINCT department_id,  
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY  
department_id) AS median_salary  
FROM employees;
```

57. Write a query to find first and last purchase date for each customer including non-buyers.

```
SELECT c.customer_id,  
MIN(s.purchase_date) AS first_purchase,  
MAX(s.purchase_date) AS last_purchase  
FROM customers c  
LEFT JOIN sales s ON c.customer_id = s.customer_id  
GROUP BY c.customer_id;
```

58. Write a query to calculate percentage difference between each month's sales and previous month.

```
WITH monthly_sales AS (  
  SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales  
  FROM sales  
  GROUP BY month  
)  
SELECT month, total_sales,  
(total_sales - LAG(total_sales) OVER (ORDER BY month)) * 100.0 /  
LAG(total_sales) OVER (ORDER BY month) AS pct_change  
FROM monthly_sales;
```

59. Write a query to find employees with longest tenure in their department.

```

WITH tenure AS (
    SELECT *, RANK() OVER (PARTITION BY department_id ORDER BY hire_date ASC) AS
    tenure_rank
    FROM employees
)
SELECT *
FROM tenure
WHERE tenure_rank = 1;

```

60. Write a query to generate sales growth percentage compared to same month last year.

```

WITH monthly_sales AS (
    SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
    FROM sales
    GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';

```

61. Write a query to identify overlapping shifts for employees.

```

SELECT s1.employee_id, s1.shift_id AS shift1, s2.shift_id AS shift2
FROM shifts s1
JOIN shifts s2 ON s1.employee_id = s2.employee_id AND s1.shift_id <> s2.shift_id
WHERE s1.start_time < s2.end_time AND s1.end_time > s2.start_time;

```

62. Write a query to calculate total revenue per customer and rank them.

```

SELECT customer_id, SUM(amount) AS total_revenue,
RANK() OVER (ORDER BY SUM(amount) DESC) AS revenue_rank
FROM sales
GROUP BY customer_id;

```

63. Write a query to find employees who never received a promotion.

```

SELECT e.*
FROM employees e
LEFT JOIN promotions p ON e.id = p.employee_id
WHERE p.employee_id IS NULL;

```

64. Write a query to find top 3 products with highest sales amount each month.

```
WITH monthly_product_sales AS (  
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS  
    total_sales  
    FROM sales  
    GROUP BY product_id, month  
)  
,  
ranked_sales AS (  
    SELECT *, RANK() OVER (PARTITION BY month ORDER BY total_sales DESC) AS  
    sales_rank  
    FROM monthly_product_sales  
)  
SELECT product_id, month, total_sales  
FROM ranked_sales  
WHERE sales_rank <= 3  
ORDER BY month, sales_rank;
```

65. Write a query to find customers who placed orders only in last 30 days.

```
SELECT DISTINCT customer_id  
FROM orders  
WHERE order_date >= CURRENT_DATE - INTERVAL '30 days'  
AND customer_id NOT IN (  
    SELECT DISTINCT customer_id  
    FROM orders  
    WHERE order_date < CURRENT_DATE - INTERVAL '30 days'  
);
```

66. Write a query to find products that have never been ordered.

```
SELECT p.product_id, p.product_name  
FROM products p  
LEFT JOIN orders o ON p.product_id = o.product_id  
WHERE o.order_id IS NULL;
```

67. Write a query to calculate total sales amount and number of orders per customer in last year.

```
SELECT customer_id, COUNT(*) AS total_orders, SUM(amount) AS total_sales  
FROM sales  
WHERE sale_date >= CURRENT_DATE - INTERVAL '1 year'  
GROUP BY customer_id;
```

68. Write a query to find employees who have managed more than 3 projects.


```
SELECT manager_id, COUNT(DISTINCT project_id) AS project_count
FROM projects
GROUP BY manager_id
HAVING COUNT(DISTINCT project_id) > 3;
```

♦ Employee & Manager Queries

97. Write a query to calculate the difference in days between each employee's hire date and their manager's hire date.

```
SELECT e.name AS employee, m.name AS manager,
DATEDIFF(day, m.hire_date, e.hire_date) AS days_difference
FROM employees e
JOIN employees m ON e.manager_id = m.id;
```

98. Write a query to find the department with the highest average years of experience.

```
SELECT department_id, AVG(EXTRACT(year FROM CURRENT_DATE - hire_date)) AS
avg_experience_years
FROM employees
GROUP BY department_id
ORDER BY avg_experience_years DESC
LIMIT 1;
```

99. Write a query to identify employees with overlapping project assignments.

```
SELECT p1.employee_id, p1.project_id AS project1, p2.project_id AS project2
FROM project_assignments p1
JOIN project_assignments p2 ON p1.employee_id = p2.employee_id AND p1.project_id
<> p2.project_id
WHERE p1.start_date < p2.end_date AND p1.end_date > p2.start_date;
```

100. Write a query to find customers who made purchases in every month of the current year.

```
WITH months AS (SELECT generate_series(1, 12) AS month),
customer_months AS (
  SELECT customer_id, EXTRACT(MONTH FROM purchase_date) AS month
  FROM sales
  WHERE EXTRACT(YEAR FROM purchase_date) = EXTRACT(YEAR FROM CURRENT_DATE)
  GROUP BY customer_id, EXTRACT(MONTH FROM purchase_date)
)
SELECT customer_id
FROM customer_months
GROUP BY customer_id
HAVING COUNT(DISTINCT month) = 12;
```

101. Write a query to list employees who earn more than all their subordinates.

```
SELECT e.id, e.name, e.salary
FROM employees e
WHERE e.salary > ALL (SELECT salary FROM employees sub WHERE sub.manager_id = e.id);
```

102. Write a query to get the product with the highest sales for each category.

```
WITH category_sales AS (
  SELECT category_id, product_id, SUM(amount) AS total_sales,
  RANK() OVER (PARTITION BY category_id ORDER BY SUM(amount) DESC) AS sales_rank
  FROM sales
  GROUP BY category_id, product_id
)
SELECT category_id, product_id, total_sales
FROM category_sales
WHERE sales_rank = 1;
```

103. Write a query to find customers who haven't ordered in the last 6 months.

```
SELECT c.customer_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months' OR
MAX(o.order_date) IS NULL;
```

104. Write a query to find maximum salary gap between employees in same department.

```
SELECT department_id, MAX(salary) - MIN(salary) AS salary_gap
FROM employees
GROUP BY department_id;
```

105. Write a recursive query to compute total budget under each manager.

```
WITH RECURSIVE manager_budget AS (
  SELECT id, manager_id, budget
  FROM departments
  UNION ALL
  SELECT d.id, d.manager_id, mb.budget
  FROM departments d
  JOIN manager_budget mb ON d.manager_id = mb.id
)
SELECT manager_id, SUM(budget) AS total_budget
FROM manager_budget
GROUP BY manager_id;
```

106. Write a query to detect gaps in invoice numbers.

```
WITH numbered_invoices AS (
  SELECT invoice_number, ROW_NUMBER() OVER (ORDER BY invoice_number) AS rn
  FROM invoices
)
SELECT invoice_number + 1 AS missing_invoice
FROM numbered_invoices ni
WHERE (invoice_number + 1) <> (
  SELECT invoice_number FROM numbered_invoices WHERE rn = ni.rn + 1
);
```

107. Write a query to rank employees by salary within department but restart rank every 10 employees.

```
WITH ranked_employees AS (
  SELECT e.*, ROW_NUMBER() OVER (PARTITION BY department_id ORDER BY salary
  DESC) AS rn
  FROM employees e
)
SELECT *, ((rn-1)/10)+1 AS rank_group
FROM ranked_employees;
```

108. Write a query to calculate moving median of daily sales over last 7 days per product.

```
WITH daily_sales AS (  
    SELECT product_id, sale_date, SUM(amount) AS total_sales  
    FROM sales  
    GROUP BY product_id, sale_date  
)  
SELECT product_id, sale_date,  
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)  
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND  
CURRENT ROW) AS moving_median  
FROM daily_sales;
```

109. Write a query to find customers who purchased both product A and product B.

```
SELECT customer_id  
FROM sales  
WHERE product_id IN ('A', 'B')  
GROUP BY customer_id  
HAVING COUNT(DISTINCT product_id) = 2;
```

◆ Calendar, Attendance & Hierarchy

111. Write a query to find employees who worked in more than 3 different departments.

```
SELECT employee_id  
FROM employee_department_history  
GROUP BY employee_id  
HAVING COUNT(DISTINCT department_id) > 3;
```

112. Write a query to calculate percentage contribution of each product's sales to total monthly sales.

```

WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
product_sales
    FROM sales
    GROUP BY product_id, month
),
total_monthly_sales AS (
    SELECT month, SUM(product_sales) AS total_sales
    FROM monthly_sales
    GROUP BY month
)
SELECT m.product_id, m.month,
(m.product_sales * 100.0 / t.total_sales) AS pct_contribution
FROM monthly_sales m
JOIN total_monthly_sales t ON m.month = t.month;

```

113. Write a query to identify gaps and islands in attendance records.

```

WITH attendance_groups AS (
    SELECT employee_id, attendance_date,
attendance_date - INTERVAL '1 day' * ROW_NUMBER() OVER (PARTITION BY
employee_id ORDER BY attendance_date) AS grp
    FROM attendance
)
SELECT employee_id, MIN(attendance_date) AS start_date, MAX(attendance_date) AS
end_date, COUNT(*) AS consecutive_days
FROM attendance_groups
GROUP BY employee_id, grp
ORDER BY employee_id, start_date;

```

114. Write a recursive query to list all descendants of a manager.

```

WITH RECURSIVE descendants AS (
    SELECT id, name, manager_id
    FROM employees
    WHERE manager_id = 100
    UNION ALL
    SELECT e.id, e.name, e.manager_id
    FROM employees e
    INNER JOIN descendants d ON e.manager_id = d.id
)
SELECT * FROM descendants;

```

115. Write a query to calculate a 3-month moving average of monthly sales per product.

```

WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;

```

116. Write a query to find employees who have the same hire date as their managers.

```

SELECT e.name AS employee_name, m.name AS manager_name, e.hire_date
FROM employees e
JOIN employees m ON e.manager_id = m.id
WHERE e.hire_date = m.hire_date;

```

117. Write a query to find products with increasing sales over last 3 months.

```

WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
),
ranked_sales AS (
    SELECT product_id, month, total_sales,
ROW_NUMBER() OVER (PARTITION BY product_id ORDER BY month DESC) AS rn
    FROM monthly_sales
)
SELECT ms1.product_id
FROM ranked_sales ms1
JOIN ranked_sales ms2 ON ms1.product_id = ms2.product_id AND ms1.rn = 1 AND
ms2.rn = 2
JOIN ranked_sales ms3 ON ms1.product_id = ms3.product_id AND ms3.rn = 3
WHERE ms3.total_sales < ms2.total_sales AND ms2.total_sales < ms1.total_sales;

```

118. Write a query to get the Nth highest salary per department.

```
SELECT department_id, salary
FROM (
    SELECT department_id, salary, ROW_NUMBER() OVER (PARTITION BY department_id
    ORDER BY salary DESC) AS rn
    FROM employees
) sub
WHERE rn = N;
```

119. Write a query to find customers who made purchases in every category available.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM
sales);
```

120. Write a query to calculate average tenure of employees by department.

```
SELECT department_id, AVG(DATE_PART('year', CURRENT_DATE - hire_date)) AS
avg_tenure_years
FROM employees
GROUP BY department_id;
```

◆ Sales & Orders

121. Write a query to find customers who purchased more than once in the same day.

```
SELECT customer_id, purchase_date, COUNT(*) AS purchase_count
FROM sales
GROUP BY customer_id, purchase_date
HAVING COUNT(*) > 1;
```

122. Write a query to find customers who purchased all products in a category.


```

SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);

```

123. Write a query to find customers who purchased both product A and product B.

```

SELECT customer_id
FROM sales
WHERE product_id IN ('A', 'B')
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = 2;

```

124. Write a query to calculate cumulative percentage of total sales per product.

```

SELECT product_id, sale_amount,
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)
OVER () AS cumulative_pct
FROM sales;

```

125. Write a query to calculate monthly sales growth compared to same month last year.

```

WITH monthly_sales AS (
    SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
    FROM sales
    GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';

```

126. Write a query to calculate moving median of daily sales over last 7 days per product.

```
WITH daily_sales AS (
  SELECT product_id, sale_date, SUM(amount) AS total_sales
  FROM sales
  GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;
```

127. Write a query to calculate 3-month moving average of monthly sales per product.

```
WITH monthly_sales AS (
  SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
  FROM sales
  GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;
```

128. Write a query to calculate percentage contribution of each product's sales to total monthly sales.

```
WITH monthly_sales AS (
  SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
product_sales
  FROM sales
  GROUP BY product_id, month
),
total_monthly_sales AS (
  SELECT month, SUM(product_sales) AS total_sales
  FROM monthly_sales
  GROUP BY month
)
SELECT m.product_id, m.month,
(m.product_sales * 100.0 / t.total_sales) AS pct_contribution
FROM monthly_sales m
JOIN total_monthly_sales t ON m.month = t.month;
```

◆ Employee & Department Analysis

161. Write a query to find employees who have worked in multiple departments over time.

```
SELECT employee_id
FROM employee_department_history
GROUP BY employee_id
HAVING COUNT(DISTINCT department_id) > 1;
```

162. Write a query to find employees who have worked in more than 3 different departments.

```
SELECT employee_id
FROM employee_department_history
GROUP BY employee_id
HAVING COUNT(DISTINCT department_id) > 3;
```

163. Write a query to find employees who earn more than the average salary across the company but less than the highest salary in their department.

```
SELECT *
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees)
AND salary < (SELECT MAX(salary) FROM employees WHERE department_id =
e.department_id);
```

164. Write a query to find employees who earn more than all their subordinates.

```
SELECT e.id, e.name, e.salary
FROM employees e
WHERE e.salary > ALL (SELECT salary FROM employees sub WHERE sub.manager_id =
e.id);
```

165. Write a query to find employees who have the longest tenure within their department.

```
WITH tenure AS (
    SELECT *, RANK() OVER (PARTITION BY department_id ORDER BY hire_date ASC) AS
tenure_rank
    FROM employees
)
SELECT *
FROM tenure
WHERE tenure_rank = 1;
```

166. Write a query to find employees who have never made a sale.

```
SELECT e.id, e.name
FROM employees e
LEFT JOIN sales s ON e.id = s.employee_id
WHERE s.sale_id IS NULL;
```

167. Write a query to find employees who have never received a promotion.

```
SELECT e.*
FROM employees e
LEFT JOIN promotions p ON e.id = p.employee_id
WHERE p.employee_id IS NULL;
```

168. Write a query to find employees with no salary changes in the last 2 years.

```
SELECT e.*
FROM employees e
LEFT JOIN salary_history sh ON e.id = sh.employee_id AND sh.change_date >=
CURRENT_DATE - INTERVAL '2 years'
WHERE sh.employee_id IS NULL;
```

169. Write a query to find employees who haven't received a raise in more than a year.

```
SELECT e.name
FROM employees e
JOIN salary_history sh ON e.id = sh.employee_id
GROUP BY e.id, e.name
HAVING MAX(sh.raise_date) < CURRENT_DATE - INTERVAL '1 year';
```

170. Write a query to find employees who don't have a department assigned.

```
SELECT *
FROM employees
WHERE department_id IS NULL;
```

◆ Customer & Sales Analysis

171. Write a query to find customers who have not made any purchase.

```
SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN sales s ON c.customer_id = s.customer_id
WHERE s.sale_id IS NULL;
```

172. Write a query to find customers who purchased more than once in the same day.

```
SELECT customer_id, purchase_date, COUNT(*) AS purchase_count
FROM sales
GROUP BY customer_id, purchase_date
HAVING COUNT(*) > 1;
```

173. Write a query to find customers who purchased all products in a specific category.

```
SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);
```

174. Write a query to find customers who purchased both product A and product B.

```
SELECT customer_id
FROM sales
WHERE product_id IN ('A', 'B')
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = 2;
```

175. Write a query to find customers who placed orders only in the last 30 days.

```
SELECT DISTINCT customer_id
FROM orders
WHERE order_date >= CURRENT_DATE - INTERVAL '30 days'
AND customer_id NOT IN (
    SELECT DISTINCT customer_id
    FROM orders
    WHERE order_date < CURRENT_DATE - INTERVAL '30 days'
);
```

176. Write a query to find customers who haven't ordered in the last 6 months.

```
SELECT c.customer_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months' OR
MAX(o.order_date) IS NULL;
```

177. Write a query to find customers who made purchases in every month of the current year.

```
WITH months AS (SELECT generate_series(1, 12) AS month),
customer_months AS (
  SELECT customer_id, EXTRACT(MONTH FROM purchase_date) AS month
  FROM sales
  WHERE EXTRACT(YEAR FROM purchase_date) = EXTRACT(YEAR FROM CURRENT_DATE)
  GROUP BY customer_id, EXTRACT(MONTH FROM purchase_date)
)
SELECT customer_id
FROM customer_months
GROUP BY customer_id
HAVING COUNT(DISTINCT month) = 12;
```

178. Write a query to find customers who purchased in every category available.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM
sales);
```

179. Write a query to calculate total revenue for each customer and rank them.

```
SELECT customer_id, SUM(amount) AS total_revenue,
RANK() OVER (ORDER BY SUM(amount) DESC) AS revenue_rank
FROM sales
GROUP BY customer_id;
```

180. Write a query to calculate total sales amount and number of orders per customer in the last year.

```
SELECT customer_id, COUNT(*) AS total_orders, SUM(amount) AS total_sales
FROM sales
WHERE sale_date >= CURRENT_DATE - INTERVAL '1 year'
GROUP BY customer_id;
```

◆ Advanced Analytics

181. Write a query to calculate cumulative distribution (CDF) of salaries.

```
SELECT name, salary,
CUME_DIST() OVER (ORDER BY salary) AS salary_cdf
FROM employees;
```

182. Write a query to calculate cumulative percentage of total sales per product.

```
SELECT product_id, sale_amount,
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)
OVER () AS cumulative_pct
FROM sales;
```

183. Write a query to calculate monthly sales growth compared to same month last year.

```
WITH monthly_sales AS (
  SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
  FROM sales
  GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';
```

184. Write a query to calculate percentage change in sales compared to previous month for each product.


```

SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
AS total_sales
    FROM sales
    GROUP BY product_id, sale_month
) monthly_sales;

```

185. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```

SELECT name, hire_date, salary,
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg_salary
FROM employees;

```

186. Write a query to calculate 3-month moving average of monthly sales per product.

```

WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;

```

187. Write a query to calculate moving median of daily sales over last 7 days per product.

```

WITH daily_sales AS (
  SELECT product_id, sale_date, SUM(amount) AS total_sales
  FROM sales
  GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;

```

◆ Employee & Hierarchy

188. Write a query to find employees who are at the lowest level in the hierarchy (no subordinates).

```

SELECT *
FROM employees e
WHERE NOT EXISTS (SELECT 1 FROM employees sub WHERE sub.manager_id = e.id);

```

189. Write a query to calculate hierarchical depth of each employee.

```

WITH RECURSIVE employee_depth AS (
  SELECT id, name, manager_id, 1 AS depth
  FROM employees
  WHERE manager_id IS NULL
  UNION ALL
  SELECT e.id, e.name, e.manager_id, ed.depth + 1
  FROM employees e
  JOIN employee_depth ed ON e.manager_id = ed.id
)
SELECT * FROM employee_depth;

```

190. Write a query to list all ancestors (managers) of a given employee.

```

WITH RECURSIVE ancestors AS (
  SELECT id, name, manager_id
  FROM employees
  WHERE id = 123
  UNION ALL
  SELECT e.id, e.name, e.manager_id
  FROM employees e
  JOIN ancestors a ON e.id = a.manager_id
)
SELECT * FROM ancestors WHERE id != 123;

```

191. Write a query to list all descendants of a manager.

```
WITH RECURSIVE descendants AS (  
  SELECT id, name, manager_id  
  FROM employees  
  WHERE manager_id = 100  
  UNION ALL  
  SELECT e.id, e.name, e.manager_id  
  FROM employees e  
  INNER JOIN descendants d ON e.manager_id = d.id  
)  
SELECT * FROM descendants;
```

192. Write a query to detect circular references in employee-manager hierarchy.

```
WITH RECURSIVE mgr_path (id, manager_id, path) AS (  
  SELECT id, manager_id, ARRAY[id]  
  FROM employees  
  WHERE manager_id IS NOT NULL  
  UNION ALL  
  SELECT e.id, e.manager_id, path || e.id  
  FROM employees e  
  JOIN mgr_path mp ON e.manager_id = mp.id  
  WHERE NOT e.id = ANY(path)  
)  
SELECT DISTINCT id  
FROM mgr_path  
WHERE id = ANY(path);
```

◆ Sales & Orders**193. Write a query to find the most recent purchase per customer.**

```
SELECT *  
FROM (  
  SELECT *, ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY purchase_date  
  DESC) AS rn  
  FROM sales  
) sub  
WHERE rn = 1;
```

194. Write a query to find the second most recent order date per customer.

```

SELECT customer_id, order_date
FROM (
    SELECT customer_id, order_date,
    ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date DESC) AS rn
    FROM orders
) sub
WHERE rn = 2;

```

195. Write a query to retrieve the last 5 orders for each customer.

```

SELECT *
FROM (
    SELECT o.*, ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date
DESC) AS rn
    FROM orders o
) sub
WHERE rn <= 5;

```

196. Write a query to calculate total sales amount and number of orders per customer in the last year.

```

SELECT customer_id, COUNT(*) AS total_orders, SUM(amount) AS total_sales
FROM sales
WHERE sale_date >= CURRENT_DATE - INTERVAL '1 year'
GROUP BY customer_id;

```

197. Write a query to calculate monthly sales growth compared to same month last year.

```

WITH monthly_sales AS (
    SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
    FROM sales
    GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';

```

198. Write a query to calculate percentage change in sales compared to previous month for each product.

```

SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
AS total_sales
    FROM sales
    GROUP BY product_id, sale_month
) monthly_sales;

```

199. Write a query to rank salespeople by monthly sales, resetting rank every month.

```

SELECT salesperson_id, sale_month, total_sales,
RANK() OVER (PARTITION BY sale_month ORDER BY total_sales DESC) AS monthly_rank
FROM (
    SELECT salesperson_id, DATE_TRUNC('month', sale_date) AS sale_month,
SUM(amount) AS total_sales
    FROM sales
    GROUP BY salesperson_id, sale_month
) monthly_sales;

```

200. Write a query to find top 3 products with highest sales each month.

```

WITH monthly_product_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
),
ranked_sales AS (
    SELECT *, RANK() OVER (PARTITION BY month ORDER BY total_sales DESC) AS
sales_rank
    FROM monthly_product_sales
)
SELECT product_id, month, total_sales
FROM ranked_sales
WHERE sales_rank <= 3
ORDER BY month, sales_rank;

```

◆ Advanced Analytics

201. Write a query to calculate cumulative distribution (CDF) of salaries.

```
SELECT name, salary,  
CUME_DIST() OVER (ORDER BY salary) AS salary_cdf  
FROM employees;
```

202. Write a query to calculate cumulative percentage of total sales per product.

```
SELECT product_id, sale_amount,  
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)  
OVER () AS cumulative_pct  
FROM sales;
```

203. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,  
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)  
AS moving_avg_salary  
FROM employees;
```

204. Write a query to calculate 3-month moving average of monthly sales per product.

```
WITH monthly_sales AS (  
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS  
    total_sales  
    FROM sales  
    GROUP BY product_id, month  
)  
SELECT product_id, month, total_sales,  
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2  
PRECEDING AND CURRENT ROW) AS moving_avg  
FROM monthly_sales;
```

205. Write a query to calculate moving median of daily sales over last 7 days per product.

```
WITH daily_sales AS (
  SELECT product_id, sale_date, SUM(amount) AS total_sales
  FROM sales
  GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;
```

206. Write a query to calculate salary percentile (e.g., 90th) per department.

```
SELECT department_id, salary,
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY
department_id) AS pct_90_salary
FROM employees;
```

207. Write a query to rank employees by salary within their department and calculate percent rank.

```
SELECT name, department_id, salary,
RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS salary_rank,
PERCENT_RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS
salary_percent_rank
FROM employees;
```

208. Write a query to calculate difference between current and previous sales partitioned by product.

```
SELECT product_id, sale_date, amount,
amount - LAG(amount) OVER (PARTITION BY product_id ORDER BY sale_date) AS
sales_diff
FROM sales;
```

209. Write a query to calculate difference between current row and previous row's salary.

```
SELECT name, salary,
salary - LAG(salary) OVER (ORDER BY id) AS salary_diff
FROM employees;
```

210. Write a query to calculate difference in days between employee's hire date and manager's hire date.


```
SELECT e.name AS employee, m.name AS manager,
DATEDIFF(day, m.hire_date, e.hire_date) AS days_difference
FROM employees e
JOIN employees m ON e.manager_id = m.id;
```

◆ Employee & Salary Analysis

238. Write a query to find employees whose salary is above their department's average but below the overall company average.

```
SELECT *
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id =
e.department_id)
AND salary < (SELECT AVG(salary) FROM employees);
```

239. Write a query to find employees whose salary is a prime number.

```
WITH primes AS (
  SELECT generate_series(2, (SELECT MAX(salary) FROM employees)) AS num
  EXCEPT
  SELECT num FROM (
    SELECT num, UNNEST(ARRAY(
      SELECT generate_series(2, FLOOR(SQRT(num))) AS divisor
    )) AS divisor
    WHERE num % divisor = 0
  ) composite
)
SELECT *
FROM employees
WHERE salary IN (SELECT num FROM primes);
```

240. Write a query to find employees with salary in the top 10% in their department.

```
SELECT *
FROM (
  SELECT e.*, NTILE(10) OVER (PARTITION BY department_id ORDER BY salary DESC)
  AS decile
  FROM employees e
) sub
WHERE decile = 1;
```

241. Write a query to find employees whose names start and end with the same letter.

```
SELECT *  
FROM employees  
WHERE LEFT(name, 1) = RIGHT(name, 1);
```

242. Write a query to find employees who joined in the last 6 months.

```
SELECT *  
FROM employees  
WHERE join_date >= CURRENT_DATE - INTERVAL '6 months';
```

243. Write a query to find employees who have the same hire date as their managers.

```
SELECT e.name AS employee_name, m.name AS manager_name, e.hire_date  
FROM employees e  
JOIN employees m ON e.manager_id = m.id  
WHERE e.hire_date = m.hire_date;
```

244. Write a query to find employees who earn more than their manager.

```
SELECT e.name AS Employee, e.salary, m.name AS Manager, m.salary AS  
ManagerSalary  
FROM employees e  
JOIN employees m ON e.manager_id = m.id  
WHERE e.salary > m.salary;
```

245. Write a query to find employees who earn the same salary as their manager.

```
SELECT e.name AS Employee, e.salary, m.name AS Manager  
FROM employees e  
JOIN employees m ON e.manager_id = m.id  
WHERE e.salary = m.salary;
```

246. Write a query to find employees who have worked in multiple departments.

```
SELECT employee_id  
FROM employee_department_history  
GROUP BY employee_id  
HAVING COUNT(DISTINCT department_id) > 1;
```

247. Write a query to find employees who have managed more than 3 projects.

```
SELECT manager_id, COUNT(DISTINCT project_id) AS project_count
FROM projects
GROUP BY manager_id
HAVING COUNT(DISTINCT project_id) > 3;
```

♦ Customer & Orders

248. Write a query to find customers who purchased all products in a specific category.

```
SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);
```

249. Write a query to find customers who purchased both product A and product B.

```
SELECT customer_id
FROM sales
WHERE product_id IN ('A', 'B')
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = 2;
```

250. Write a query to find customers who purchased in every category available.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM sales);
```

251. Write a query to find customers who purchased more than once in the same day.

```
SELECT customer_id, purchase_date, COUNT(*) AS purchase_count
FROM sales
GROUP BY customer_id, purchase_date
HAVING COUNT(*) > 1;
```

252. Write a query to find customers who haven't ordered in the last 6 months.

```
SELECT c.customer_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months' OR
MAX(o.order_date) IS NULL;
```

253. Write a query to find customers who placed orders only in the last 30 days.

```
SELECT DISTINCT customer_id
FROM orders
WHERE order_date >= CURRENT_DATE - INTERVAL '30 days'
AND customer_id NOT IN (
    SELECT DISTINCT customer_id
    FROM orders
    WHERE order_date < CURRENT_DATE - INTERVAL '30 days'
);
```

254. Write a query to find customers who made purchases in every month of the current year.

```
WITH months AS (SELECT generate_series(1, 12) AS month),
customer_months AS (
    SELECT customer_id, EXTRACT(MONTH FROM purchase_date) AS month
    FROM sales
    WHERE EXTRACT(YEAR FROM purchase_date) = EXTRACT(YEAR FROM CURRENT_DATE)
    GROUP BY customer_id, EXTRACT(MONTH FROM purchase_date)
)
SELECT customer_id
FROM customer_months
GROUP BY customer_id
HAVING COUNT(DISTINCT month) = 12;
```

255. Write a query to find customers who have not made any purchase.

```
SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN sales s ON c.customer_id = s.customer_id
WHERE s.sale_id IS NULL;
```

◆ Advanced Analytics

256. Write a query to calculate cumulative distribution (CDF) of salaries.

```
SELECT name, salary,
CUME_DIST() OVER (ORDER BY salary) AS salary_cdf
FROM employees;
```

257. Write a query to calculate cumulative percentage of total sales per product.

```
SELECT product_id, sale_amount,
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)
OVER () AS cumulative_pct
FROM sales;
```

258. Write a query to calculate monthly sales growth compared to same month last year.

```
WITH monthly_sales AS (
  SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
  FROM sales
  GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';
```

259. Write a query to calculate percentage change in sales compared to previous month for each product.

```
SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
  SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
  AS total_sales
  FROM sales
  GROUP BY product_id, sale_month
) monthly_sales;
```

260. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg_salary
FROM employees;
```

261. Write a query to calculate 3-month moving average of monthly sales per product.

```
WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;
```

262. Write a query to calculate moving median of daily sales over last 7 days per product.

```
WITH daily_sales AS (
    SELECT product_id, sale_date, SUM(amount) AS total_sales
    FROM sales
    GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;
```

◆ Employee & Department Analysis

288. Write a query to find the department with the highest average salary.

```
WITH avg_salaries AS (
    SELECT department_id, AVG(salary) AS avg_salary
    FROM employees
    GROUP BY department_id
)
SELECT *
FROM avg_salaries
WHERE avg_salary = (SELECT MAX(avg_salary) FROM avg_salaries);
```

289. Write a query to find the department with the lowest average salary.

```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
ORDER BY avg_salary
LIMIT 1;
```

290. Write a query to find the department with the highest number of employees.

```
SELECT department_id, COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
ORDER BY employee_count DESC
LIMIT 1;
```

291. Write a query to find the maximum salary gap between employees in the same department.

```
SELECT department_id, MAX(salary) - MIN(salary) AS salary_gap
FROM employees
GROUP BY department_id;
```

292. Write a query to calculate average tenure of employees by department.

```
SELECT department_id, AVG(DATE_PART('year', CURRENT_DATE - hire_date)) AS
avg_tenure_years
FROM employees
GROUP BY department_id;
```

293. Write a query to calculate average years of experience by department.

```
SELECT department_id, AVG(EXTRACT(year FROM CURRENT_DATE - hire_date)) AS
avg_experience_years
FROM employees
GROUP BY department_id
ORDER BY avg_experience_years DESC;
```

294. Write a query to find employees with longest tenure in their department.


```
WITH tenure AS (  
    SELECT *, RANK() OVER (PARTITION BY department_id ORDER BY hire_date ASC) AS  
    tenure_rank  
    FROM employees  
)  
SELECT *  
FROM tenure  
WHERE tenure_rank = 1;
```

295. Write a query to find employees who joined in each year and calculate running total hires.

```
SELECT join_year, COUNT(*) AS yearly_hires,  
SUM(COUNT(*)) OVER (ORDER BY join_year) AS running_total_hires  
FROM (  
    SELECT EXTRACT(YEAR FROM hire_date) AS join_year  
    FROM employees  
) sub  
GROUP BY join_year  
ORDER BY join_year;
```

◆ Customer & Orders

296. Write a query to find first and last purchase date for each customer.

```
SELECT customer_id,  
MIN(purchase_date) AS first_purchase,  
MAX(purchase_date) AS last_purchase  
FROM sales  
GROUP BY customer_id;
```

297. Write a query to find first and last purchase date including customers who never purchased.

```
SELECT c.customer_id,  
MIN(s.purchase_date) AS first_purchase,  
MAX(s.purchase_date) AS last_purchase  
FROM customers c  
LEFT JOIN sales s ON c.customer_id = s.customer_id  
GROUP BY c.customer_id;
```

298. Write a query to find customers who purchased more than once in the same day.

```
SELECT customer_id, purchase_date, COUNT(*) AS purchase_count
FROM sales
GROUP BY customer_id, purchase_date
HAVING COUNT(*) > 1;
```

299. Write a query to find customers who purchased in every category available.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM
sales);
```

300. Write a query to find customers who purchased all products in a specific category.

```
SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);
```

301. Write a query to find customers who purchased both product A and product B.

```
SELECT customer_id
FROM sales
WHERE product_id IN ('A', 'B')
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = 2;
```

302. Write a query to find customers who placed orders only in the last 30 days.

```

SELECT DISTINCT customer_id
FROM orders
WHERE order_date >= CURRENT_DATE - INTERVAL '30 days'
AND customer_id NOT IN (
    SELECT DISTINCT customer_id
    FROM orders
    WHERE order_date < CURRENT_DATE - INTERVAL '30 days'
);

```

303. Write a query to find customers who haven't ordered in the last 6 months.

```

SELECT c.customer_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months' OR
MAX(o.order_date) IS NULL;

```

304. Write a query to find customers who made purchases in every month of the current year.

```

WITH months AS (SELECT generate_series(1, 12) AS month),
customer_months AS (
    SELECT customer_id, EXTRACT(MONTH FROM purchase_date) AS month
    FROM sales
    WHERE EXTRACT(YEAR FROM purchase_date) = EXTRACT(YEAR FROM CURRENT_DATE)
    GROUP BY customer_id, EXTRACT(MONTH FROM purchase_date)
)
SELECT customer_id
FROM customer_months
GROUP BY customer_id
HAVING COUNT(DISTINCT month) = 12;

```

305. Write a query to find customers who have not made any purchase.

```

SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN sales s ON c.customer_id = s.customer_id
WHERE s.sale_id IS NULL;

```

◆ Advanced Analytics

306. Write a query to calculate cumulative distribution (CDF) of salaries.

```
SELECT name, salary,
CUME_DIST() OVER (ORDER BY salary) AS salary_cdf
FROM employees;
```

307. Write a query to calculate cumulative percentage of total sales per product.

```
SELECT product_id, sale_amount,
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)
OVER () AS cumulative_pct
FROM sales;
```

308. Write a query to calculate monthly sales growth compared to same month last year.

```
WITH monthly_sales AS (
  SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
  FROM sales
  GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';
```

309. Write a query to calculate percentage change in sales compared to previous month for each product.

```
SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
  SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
AS total_sales
  FROM sales
  GROUP BY product_id, sale_month
) monthly_sales;
```

310. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg_salary
FROM employees;
```

311. Write a query to calculate 3-month moving average of monthly sales per product.

```
WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;
```

312. Write a query to calculate moving median of daily sales over last 7 days per product.

```
WITH daily_sales AS (
    SELECT product_id, sale_date, SUM(amount) AS total_sales
    FROM sales
    GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;
```

313. Write a query to calculate salary percentile (e.g., 90th) per department.

```
SELECT department_id, salary,
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY
department_id) AS pct_90_salary
FROM employees;
```

◆ Employee & Salary Analysis

338. Write a query to find the second highest salary.

```
SELECT MAX(salary) AS SecondHighestSalary
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

339. Write a query to find the Nth highest salary.

```
SELECT DISTINCT salary
FROM employees
ORDER BY salary DESC
LIMIT 1 OFFSET N-1;
```

340. Write a query to rank employees by salary with ties handled properly.

```
SELECT name, salary,
RANK() OVER (ORDER BY salary DESC) AS salary_rank
FROM employees;
```

341. Write a query to calculate the median salary.

```
SELECT AVG(salary) AS median_salary
FROM (
    SELECT salary
    FROM employees
    ORDER BY salary
    LIMIT 2 - (SELECT COUNT(*) FROM employees) % 2
    OFFSET (SELECT (COUNT(*) - 1) / 2 FROM employees)
) AS median_subquery;
```

342. Write a query to calculate the moving average of salaries over the last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg_salary
FROM employees;
```

343. Write a query to calculate the difference between current row and previous row's salary.

```
SELECT name, salary,  
salary - LAG(salary) OVER (ORDER BY id) AS salary_diff  
FROM employees;
```

344. Write a query to calculate cumulative distribution (CDF) of salaries.

```
SELECT name, salary,  
CUME_DIST() OVER (ORDER BY salary) AS salary_cdf  
FROM employees;
```

345. Write a query to calculate salary percentile (e.g., 90th) per department.

```
SELECT department_id, salary,  
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY  
department_id) AS pct_90_salary  
FROM employees;
```

346. Write a query to calculate salary difference between employee and manager hire dates.

```
SELECT e.name AS employee, m.name AS manager,  
DATEDIFF(day, m.hire_date, e.hire_date) AS days_difference  
FROM employees e  
JOIN employees m ON e.manager_id = m.id;
```

◆ Customer & Orders

347. Write a query to find customers who have not made any purchase.

```
SELECT c.customer_id, c.name  
FROM customers c  
LEFT JOIN sales s ON c.customer_id = s.customer_id  
WHERE s.sale_id IS NULL;
```

348. Write a query to find customers who purchased more than once in the same day.

```
SELECT customer_id, purchase_date, COUNT(*) AS purchase_count  
FROM sales  
GROUP BY customer_id, purchase_date  
HAVING COUNT(*) > 1;
```


349. Write a query to find customers who purchased all products in a specific category.

```
SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);
```

350. Write a query to find customers who purchased both product A and product B.

```
SELECT customer_id
FROM sales
WHERE product_id IN ('A', 'B')
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = 2;
```

351. Write a query to find customers who purchased in every category available.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM sales);
```

352. Write a query to find customers who placed orders only in the last 30 days.

```
SELECT DISTINCT customer_id
FROM orders
WHERE order_date >= CURRENT_DATE - INTERVAL '30 days'
AND customer_id NOT IN (
    SELECT DISTINCT customer_id
    FROM orders
    WHERE order_date < CURRENT_DATE - INTERVAL '30 days'
);
```

353. Write a query to find customers who haven't ordered in the last 6 months.

```

SELECT c.customer_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months' OR
MAX(o.order_date) IS NULL;

```

354. Write a query to find customers who made purchases in every month of the current year.

```

WITH months AS (SELECT generate_series(1, 12) AS month),
customer_months AS (
  SELECT customer_id, EXTRACT(MONTH FROM purchase_date) AS month
  FROM sales
  WHERE EXTRACT(YEAR FROM purchase_date) = EXTRACT(YEAR FROM CURRENT_DATE)
  GROUP BY customer_id, EXTRACT(MONTH FROM purchase_date)
)
SELECT customer_id
FROM customer_months
GROUP BY customer_id
HAVING COUNT(DISTINCT month) = 12;

```

355. Write a query to find first and last purchase date for each customer.

```

SELECT customer_id,
MIN(purchase_date) AS first_purchase,
MAX(purchase_date) AS last_purchase
FROM sales
GROUP BY customer_id;

```

356. Write a query to find first and last purchase date including customers who never purchased.

```

SELECT c.customer_id,
MIN(s.purchase_date) AS first_purchase,
MAX(s.purchase_date) AS last_purchase
FROM customers c
LEFT JOIN sales s ON c.customer_id = s.customer_id
GROUP BY c.customer_id;

```

◆ Advanced Analytics

357. Write a query to calculate cumulative percentage of total sales per product.

```
SELECT product_id, sale_amount,
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)
OVER () AS cumulative_pct
FROM sales;
```

358. Write a query to calculate monthly sales growth compared to same month last year.

```
WITH monthly_sales AS (
    SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
    FROM sales
    GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';
```

359. Write a query to calculate percentage change in sales compared to previous month for each product.

```
SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
AS total_sales
    FROM sales
    GROUP BY product_id, sale_month
) monthly_sales;
```

360. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg_salary
FROM employees;
```

361. Write a query to calculate 3-month moving average of monthly sales per product.

```

WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;

```

362. Write a query to calculate moving median of daily sales over last 7 days per product.

```

WITH daily_sales AS (
    SELECT product_id, sale_date, SUM(amount) AS total_sales
    FROM sales
    GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;

```

363. Write a query to calculate salary percentile (e.g., 90th) per department.

```

SELECT department_id, salary,
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY
department_id) AS pct_90_salary
FROM employees;

```

◆ Employee & Department Analysis

388. Write a query to count employees in each department having more than 5 employees.

```

SELECT department_id, COUNT(*) AS num_employees
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;

```

389. Write a query to find departments with no employees.

```
SELECT d.department_name
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
WHERE e.id IS NULL;
```

390. Write a query to list all departments and their employee counts, including departments with zero employees.

```
SELECT d.department_id, d.department_name, COUNT(e.id) AS employee_count
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name;
```

391. Write a query to find employees without a department assigned.

```
SELECT *
FROM employees
WHERE department_id IS NULL;
```

392. Write a query to find employees who earn more than the average salary in their department but less than the company-wide average.

```
SELECT *
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id =
e.department_id)
AND salary < (SELECT AVG(salary) FROM employees);
```

393. Write a query to find employees who earn more than the average salary across the company but less than the highest salary in their department.

```
SELECT *
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees)
AND salary < (SELECT MAX(salary) FROM employees WHERE department_id =
e.department_id);
```

394. Write a query to find employees with salary above the average salary of their department but below the company-wide average.

```
SELECT *
FROM employees e
WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id =
e.department_id)
AND salary < (SELECT AVG(salary) FROM employees);
```

395. Write a query to find employees who earn more than all their subordinates.

```
SELECT e.id, e.name, e.salary
FROM employees e
WHERE e.salary > ALL (SELECT salary FROM employees sub WHERE sub.manager_id =
e.id);
```

396. Write a query to find employees who have worked in more than 3 different departments.

```
SELECT employee_id
FROM employee_department_history
GROUP BY employee_id
HAVING COUNT(DISTINCT department_id) > 3;
```

◆ Customer & Orders

397. Write a query to find duplicate rows based on multiple columns.

```
SELECT column1, column2, COUNT(*)
FROM table_name
GROUP BY column1, column2
HAVING COUNT(*) > 1;
```

398. Write a query to find customers who purchased more than once in the same day.

```
SELECT customer_id, purchase_date, COUNT(*) AS purchase_count
FROM sales
GROUP BY customer_id, purchase_date
HAVING COUNT(*) > 1;
```

399. Write a query to find customers who purchased all products in a specific category.

```
SELECT customer_id
FROM sales
WHERE category_id = 10
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = (
    SELECT COUNT(DISTINCT product_id)
    FROM products
    WHERE category_id = 10
);
```

400. Write a query to find customers who purchased both product A and product B.

```
SELECT customer_id
FROM sales
WHERE product_id IN ('A', 'B')
GROUP BY customer_id
HAVING COUNT(DISTINCT product_id) = 2;
```

401. Write a query to find customers who purchased in every category available.

```
SELECT customer_id
FROM sales
GROUP BY customer_id
HAVING COUNT(DISTINCT category_id) = (SELECT COUNT(DISTINCT category_id) FROM sales);
```

402. Write a query to find customers who placed orders only in the last 30 days.

```
SELECT DISTINCT customer_id
FROM orders
WHERE order_date >= CURRENT_DATE - INTERVAL '30 days'
AND customer_id NOT IN (
    SELECT DISTINCT customer_id
    FROM orders
    WHERE order_date < CURRENT_DATE - INTERVAL '30 days'
);
```

403. Write a query to find customers who haven't ordered in the last 6 months.


```
SELECT c.customer_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_id
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months' OR
MAX(o.order_date) IS NULL;
```

404. Write a query to find customers who made purchases in every month of the current year.

```
WITH months AS (SELECT generate_series(1, 12) AS month),
customer_months AS (
  SELECT customer_id, EXTRACT(MONTH FROM purchase_date) AS month
  FROM sales
  WHERE EXTRACT(YEAR FROM purchase_date) = EXTRACT(YEAR FROM CURRENT_DATE)
  GROUP BY customer_id, EXTRACT(MONTH FROM purchase_date)
)
SELECT customer_id
FROM customer_months
GROUP BY customer_id
HAVING COUNT(DISTINCT month) = 12;
```

405. Write a query to find customers who have not made any purchase.

```
SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN sales s ON c.customer_id = s.customer_id
WHERE s.sale_id IS NULL;
```

◆ Advanced Analytics

406. Write a query to calculate cumulative distribution (CDF) of salaries.

```
SELECT name, salary,
CUME_DIST() OVER (ORDER BY salary) AS salary_cdf
FROM employees;
```

407. Write a query to calculate cumulative percentage of total sales per product.

```
SELECT product_id, sale_amount,
SUM(sale_amount) OVER (ORDER BY sale_amount DESC) * 100.0 / SUM(sale_amount)
OVER () AS cumulative_pct
FROM sales;
```

408. Write a query to calculate monthly sales growth compared to same month last year.

```
WITH monthly_sales AS (
    SELECT DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS total_sales
    FROM sales
    GROUP BY month
)
SELECT ms1.month, ms1.total_sales,
((ms1.total_sales - ms2.total_sales) * 100.0 / ms2.total_sales) AS growth_pct
FROM monthly_sales ms1
LEFT JOIN monthly_sales ms2 ON ms1.month = ms2.month + INTERVAL '1 year';
```

409. Write a query to calculate percentage change in sales compared to previous month for each product.

```
SELECT product_id, sale_month, total_sales,
(total_sales - LAG(total_sales) OVER (PARTITION BY product_id ORDER BY
sale_month)) * 100.0 /
LAG(total_sales) OVER (PARTITION BY product_id ORDER BY sale_month) AS
pct_change
FROM (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS sale_month, SUM(amount)
AS total_sales
    FROM sales
    GROUP BY product_id, sale_month
) monthly_sales;
```

410. Write a query to calculate moving average of salaries over last 3 employees ordered by hire date.

```
SELECT name, hire_date, salary,
AVG(salary) OVER (ORDER BY hire_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg_salary
FROM employees;
```

411. Write a query to calculate 3-month moving average of monthly sales per product.

```

WITH monthly_sales AS (
    SELECT product_id, DATE_TRUNC('month', sale_date) AS month, SUM(amount) AS
total_sales
    FROM sales
    GROUP BY product_id, month
)
SELECT product_id, month, total_sales,
AVG(total_sales) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2
PRECEDING AND CURRENT ROW) AS moving_avg
FROM monthly_sales;

```

412. Write a query to calculate moving median of daily sales over last 7 days per product.

```

WITH daily_sales AS (
    SELECT product_id, sale_date, SUM(amount) AS total_sales
    FROM sales
    GROUP BY product_id, sale_date
)
SELECT product_id, sale_date,
PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY total_sales)
OVER (PARTITION BY product_id ORDER BY sale_date ROWS BETWEEN 6 PRECEDING AND
CURRENT ROW) AS moving_median
FROM daily_sales;

```

413. Write a query to calculate salary percentile (e.g., 90th) per department.

```

SELECT department_id, salary,
PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY salary) OVER (PARTITION BY
department_id) AS pct_90_salary
FROM employees;

```

414. Write a query to calculate difference between current and previous sales partitioned by product.

```

SELECT product_id, sale_date, amount,
amount - LAG(amount) OVER (PARTITION BY product_id ORDER BY sale_date) AS
sales_diff
FROM sales;

```

**